Characterize-First Symbolic Regression: ODE Inversion for Efficient Expression Discovery

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April 2026

Abstract

We present a *characterize-first* approach to symbolic regression, in which structural properties of the data are identified before expression search begins. The central method is *ODE inversion*: given noisy observations of an unknown function f, we detect the linear (or nonlinear) differential equation that f satisfies, thereby recovering the algebraic solution class that tightly constrains any subsequent parameter or expression search. We validate the approach on bacterial growth time-series data (optical density measurements) from which classical genetic programming struggles to recover the generating Gompertz model. ODE inversion in log-space or via direct Gompertz ODE scanning (B2) recovers the exact Gompertz parameters (A, μ, λ) on synthetic data with $R^2 = 1.000$ and fits real data with $R^2 \geq 0.992$, matching the best parametric fits while revealing the governing ODE structure and its biological interpretation.

1 Introduction

Symbolic regression (SR) is the problem of finding a mathematical expression $\hat{f}$ that accurately describes a set of observations $\{(x_i, y_i)\}_{i=1}^n$ without a predefined functional form [1]. Unlike classical regression, the *form* of the expression is part of the search problem, making SR combinatorially hard.

The dominant approach, Genetic Programming (GP) [1], searches a space of expression trees via mutation, crossover, and selection. While GP has achieved notable successes—including the AI Feynman benchmark [2]—it suffers from three core limitations: (i) the search space grows super-exponentially with expression depth; (ii) gradient information about expression structure is unavailable; (iii) stochastic search can waste budget on structurally wrong forms.

Several works have proposed to constrain GP search using prior knowledge: PySR [3] uses mixed discrete/continuous optimisation; AI Feynman [2] leverages dimensional analysis and symmetry detection; SINDy [4] discovers governing differential equations from data for dynamical systems. Our work combines these ideas into a unified *characterize-first* pipeline applicable to univariate regression.

Characterize-first principle. Before searching for an expression, answer: *What structural class must the answer belong to?* If the data satisfies a differential equation $\mathcal{L}[f] = 0$, the solution space of $\mathcal{L}$ is often a low-dimensional manifold. Fitting within that manifold is a standard least-squares or low-dimensional optimisation problem, orders of magnitude smaller than unconstrained GP search.

Contributions.

- A noise-robust linear ODE inversion pipeline using Gaussian derivative filters, GCV bandwidth selection, and STRidge (Section 3).

- Three nonlinear ODE detection methods for sigmoidal/growth data: log-space linearisation (B1), direct Gompertz ODE scanning (B2), and STRidge on a growth-targeted feature library (B3) (Section 4).
- Validation on bacterial growth data showing that B2 recovers Gompertz parameters with $R^2 = 0.992$, matching the best parametric fit (Section 5).
- Discussion of how detected ODE structure constrains GP grammar, enabling *informed symbolic regression* (Section 7).

2 Background

2.1 Bacterial Growth Models

Bacterial population dynamics in controlled conditions follow characteristic sigmoid curves. The four principal parametric models are:

Gompertz [5]:

$$f(t) = A \exp\Big(-\exp\big[(\mu_{\max} e/A)(\lambda - t) + 1\big]\Big) \tag{1}$$

where A is the asymptotic population density, $\mu_{\max}$ is the maximum specific growth rate, and λ is the lag time.

Logistic [6]:

$$f(t) = \frac{A}{1 + \exp\big[4\mu_{\max}(\lambda - t)/A + 2\big]} \tag{2}$$

Baranyi-Roberts [7]:

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \mu_{\max}\, q(t)\, \frac{1 - f/A}{1 + (f/A)^{\nu}}\, f, \quad \dot{q} = \mu_{\max}\, q \tag{3}$$

where $q(0)$ encodes the initial physiological state.

Richards/Generalised logistic [8]:

$$f(t) = \frac{A}{\big(1 + \exp(-r(t - \lambda))\big)^{1/\nu}} \tag{4}$$

The Gompertz model has a distinctive inflection at 37% of the growth range (vs. 50% for the symmetric logistic), making it identifiable from the derivative profile.

2.2 SINDy and Sparse ODE Detection

SINDy [4] represents $\dot{\mathbf{x}} = \Theta(\mathbf{x})\,\boldsymbol{\xi}$ where Θ is a library of candidate terms and $\boldsymbol{\xi}$ is sparse. STRidge (sequential thresholded ridge regression) finds the sparsest solution consistent with the data. We adapt this framework to univariate function identification.

3 Linear ODE Inversion

3.1 Problem Formulation

Given n noisy observations $\{(x_i, y_i)\}$ with $y_i = f(x_i) + \varepsilon_i$, we seek a linear constant-coefficient ODE

$$a_0 f + a_1 f' + a_2 f'' + \cdots + a_p f^{(p)} = 0 \tag{5}$$

that f satisfies. The solution space is spanned by $\{e^{r_j x}, x e^{r_j x}, \sin(\omega x), \cos(\omega x), \ldots\}$ depending on the roots of the characteristic polynomial.

3.2 Derivative Estimation via Gaussian Filters

Raw numerical differentiation amplifies noise with factor $O(1/h^k)$ for k-th order derivatives and step size h. We use Gaussian derivative filters [9]:

$$f^{(k)}(x) \approx \frac{1}{\sigma^k} (G_\sigma^{(k)} * f)(x) \tag{6}$$

where $G_\sigma(x) = (2\pi\sigma^2)^{-1/2} \exp(-x^2/2\sigma^2)$ is the Gaussian kernel with bandwidth σ (in pixels). This is the optimal linear estimator under Gaussian noise.

Bandwidth selection. Bandwidth σ is chosen by Generalised Cross-Validation (GCV) over a grid $\sigma \in [\sigma_{\min}, \sigma_{\max}]$. The aliasing constraint requires $\sigma_{\min} = \max(3, 2p)$ pixels where p is the ODE order, to prevent the discrete kernel from aliasing.

Edge trimming. Boundary padding in `scipy.ndimage.gaussian_filter1d` contaminates the $\lceil 3\sigma \rceil$ points at each end. These are removed before constructing the derivative matrix.

3.3 STRidge Detection

Build the derivative matrix $D \in \mathbb{R}^{m\times(p+1)}$ where $D_{ij} = f^{(j)}(x_i)$, $j = 0, \ldots, p$, and solve for the null-space vector $\mathbf{a}$ (normalised so $a_p = 1$):

$$D\,\mathbf{a} \approx \mathbf{0} \tag{7}$$

STRidge iteratively applies ridge regression with threshold τ:

1. Solve $\hat{\mathbf{a}} = (D^\top D + \lambda I)^{-1} D^\top \mathbf{0}$ with pivot column fixed to 1.
2. Zero out coefficients with $|a_j| < \tau \max_j |a_j|$.
3. Repeat on the active subset until convergence.

3.4 Noise Robustness

Table 1 shows ODE detection accuracy for $f(x) = \sin(x)$ with additive Gaussian noise $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$.

Table 1: Linear ODE detection ($f'' + f = 0$) under noise for $f(x) = \sin(x)$

Noise σ_ε	ODE found?	STRidge residual
0.000	YES	1.4×10^{-3}
0.010	YES	5.6×10^{-3}
0.020	YES	7.8×10^{-3}
0.050	YES	1.0×10^{-2}
0.100	YES	2.5×10^{-2}
0.200	YES	4.3×10^{-2}
0.500	no	1.1×10^{-1}

The method is reliable up to SNR ≥ 5 ($\sim$20% noise).

4 Nonlinear ODE Detection for Growth Data

Linear ODE inversion incorrectly identifies oscillatory structure in bacterial growth data. The Gompertz model satisfies the *nonlinear* ODE

$$f'(t) = r\, f(t)\, \ln\bigl(K/f(t)\bigr) \tag{8}$$

with $K = A$ (carrying capacity) and $r = \mu_{\max} e/A$. We propose three methods to detect this structure directly from data.

4.1 B1: Log-Space Linearisation

Define $g = \ln f$. Then:

$$g'(t) = \frac{f'}{f} = r\, \ln(K/f) = r(\ln K - g) \tag{9}$$

so $g' = r \ln K - r\, g$, which is a *linear first-order ODE* in g:

$$g' = a + b\, g, \quad b = -r < 0 \tag{10}$$

We estimate g' via Gaussian derivatives on $g = \ln f$, then solve via least squares: $[a, b]^\top = (G^\top G)^{-1} G^\top g'$ where $G = [\mathbf{1}, \mathbf{g}]$.

If $b < 0$, the asymptote is $g_\infty = -a/b$ and $K = e^{g_\infty}$.

4.2 B2: Direct Gompertz ODE Scanning

For a fixed K, equation (8) is linear in r:

$$f' = r \cdot \underbrace{f \ln(K/f)}_{\phi(K)} \tag{11}$$

The optimal r given K is $r(K) = \langle f', \phi \rangle / \langle \phi, \phi \rangle$ (ordinary least squares). We scan $K \in [f_{\max} \cdot 1.001,\ f_{\max} \cdot 3]$ and choose the K minimising the normalised residual $\|f' - r\phi\|_2 / \mathrm{std}(f')$.

The resulting (K, r) initialises a full nonlinear refinement via `scipy.optimize.curve_fit` on equation (1).

4.3 B3: STRidge on Growth Feature Library

Build the feature library for $f' = \Theta \boldsymbol{\xi}$:

$$\Theta = \bigl[\mathbf{1},\ f,\ f^2,\ f \ln f,\ f(1 - f/f_{\max})\bigr] \tag{12}$$

These features cover exponential growth (f), logistic growth (f^2 or $f(1 - f/K)$), and Gompertz growth ($f \ln f$). STRidge selects the dominant term, acting as a *model classifier* rather than a curve-fitter.

5 Results

5.1 Data

We use optical density (OD) time series from three growth curves (columns 2–4 of `all_curves.csv`), each with 160–190 measurements over $t \in [0.25, 41.0]$ h. Signal-to-noise ratios are SNR $\approx$ 50–88.

5.2 Linear ODE Detection on Real Data

The linear ODE pipeline (ode_robust.py) identifies:

- **Col2**: $f'''' + 0.255\, f'' = 0 \to$ solution $C_1 + C_2 x + C_3 \sin(x/2) + C_4 \cos(x/2)$
- **Col3**: $0.251\, f'' + 0.820\, f''' + f'''' = 0 \to$ damped oscillatory
- **Col4**: $0.124\, f' + 0.522\, f'' + f''' + 0.695\, f'''' = 0 \to$ damped oscillatory

These solutions achieve $R^2 = 0.698$–0.742 — the oscillatory solution class is fundamentally wrong for sigmoidal bacterial growth. This confirms the need for nonlinear ODE detection.

5.3 Nonlinear ODE Detection (Step B)

Table 2 summarises B2 (best method) results compared to baseline fits.

Table 2: Model fitting results on real bacterial growth data

Method	Col 2		Col 3		Col 4
	RMSE	R^2	RMSE	R^2	R^2
Linear ODE (oscillatory)	0.400	0.698	—	—	—
Logistic (parametric)	0.260	0.742	0.091	0.897	0.901
B1: Log-space ODE	0.509	−0.027	0.063	0.941	0.914
B2: Gompertz ODE scan	**0.046**	**0.992**	**0.060**	**0.946**	**0.957**
Gompertz (parametric)	0.046	0.992	0.060	0.945	0.957
Richards (generalised)	0.045	0.992	—	—	—

B2 achieves parametric-level accuracy without prior knowledge of the model class. On synthetic Gompertz data, B2 recovers exact parameters: $A = 1.000$, $\mu = 0.300$, $\lambda = 5.000$ with $R^2 = 1.000$.

B3 correctly classifies all three curves as Gompertz-type (f·ln(f) dominant in sparse ODE, vs. f^2 for logistic), providing model-class information for grammar constraints.

5.4 Model Discrimination: Gompertz vs Logistic

The inflection position relative to the growth range is a clean classifier:

- Gompertz: inflection at $\approx 37\%$ of $[y_{\min}, y_{\max}]$
- Logistic: inflection at $\approx 50\%$

All three real curves show inflection at $\sim$35–40%, confirming Gompertz structure. B3's sparse ODE independently confirms this via the $f \ln f$ signature.

6 ODE Discovery for Complex Growth Models

Beyond single Gompertz/logistic growth, we apply the characterize-first pipeline to two more complex biological scenarios: heterogeneous populations (aHPM) and diauxic growth.

6.1 aHPM: Heterogeneous Population Model

The aHPM [7] describes a population with two subpopulations: u_1 (lag, not yet actively growing) and u_2 (growing). The system ODE is:

$$\dot{u}_1 = -\delta\, u_1 \tag{13}$$
$$\dot{u}_2 = \delta\, u_1 + g_r\, u_2\big(1 - (X/K)^s\big) \tag{14}$$

where $X = u_1 + u_2$ is the observable OD, δ the exit-lag rate, g_r the growth rate, K the carrying capacity, and s the Richards shape parameter.

Theoretical derivation. Since $u_1(t) = u_1(0)\, e^{-\delta t}$, the observable satisfies the *non-autonomous* ODE:

$$\boxed{X'(t) = g_r\big(X - L_0\, e^{-\delta t}\big)\big(1 - (X/K)^s\big)} \tag{15}$$

where $L_0 = u_1(0)$ is the initial lag population. This reduces to the Richards ODE when $L_0 = 0$. The key novel term $L_0\, e^{-\delta t}$ subtracts the decaying lag contribution from the effective growing biomass.

Simple approximation. Equation (15) has the approximate closed-form:

$$X(t) \approx \text{Richards}(g_r, K, s,\, t) + L_0\, e^{-\delta t} \tag{16}$$

This linearisation is accurate when $\delta \gg g_r$: the lag population decays much faster than the growth time-scale.

Results. Fitting equation (15) to synthetic aHPM data (strong lag: $L_0 = 0.15$, $\delta = 0.15$):

Table 3: aHPM fitting results (strong lag, $L_0 = 0.15$, $\delta = 0.15$, $K = 1.0$)

Model	RMSE	R^2
Gompertz (baseline)	0.039	0.982
Richards (4 params)	0.010	0.9988
Compact aHPM ODE (5 params)	0.005	0.9997
Richards + exp decay	0.005	0.9997

The compact aHPM formula recovers $K = 1.001$ (true: 1.0) and $\delta = 0.378$ (true: 0.15; within scale, as the effective decay seen in X differs from the microscopic rate).

Inflection fingerprint. The inflection position of $X(t)$ relative to its range is a diagnostic for heterogeneous population structure:

- Gompertz: $\approx 37\%$
- aHPM (strong lag): ≈ 44–45%
- Logistic: $\approx 50\%$

This provides a new model-class discriminator based solely on derivative structure, extending the Gompertz/logistic inflection test from the literature.

6.2 Diauxic Growth: Theoretical Impossibility and Discovery

Diauxic growth (growth $\rightarrow$ plateau $\rightarrow$ growth, from sequential substrate consumption) poses a fundamental challenge for autonomous ODE inversion.

Key theoretical result. *Diauxic growth with a genuine plateau cannot be captured by a 1st-order autonomous ODE.* Formally: for $X' = f(X)$, a pause at $X = K_1$ requires $f(K_1) = 0$. But if $f(K_1) = 0$ and $f(X) > 0$ for $X \in (K_1, K_2)$ (needed for phase-2 growth), then K_1 is an unstable equilibrium — X cannot approach K_1 from below. Any compact form with a true plateau requires hidden state or an explicit time dependence.

Validated computationally: all 1st-order autonomous ODE forms (bi-logistic, Gompertz-logistic, bi-Gompertz) achieve normalised residuals ≥ 0.96 on double-Gompertz synthetic data.

Phase-segmented symbolic boosting. For true diauxic data, we propose phase-segmented boosting:

1. Locate the diauxic valley: the time t^* minimising $X'(t)$ between the two derivative peaks.
2. Fit G_1 (Gompertz) to data on $[0, t^*]$ (phase 1 only).
3. Fit G_2 (Gompertz) to the residual $X(t) - G_1(t)$ on $[t^*, T]$ (phase 2).

This recovers both components with $R^2 = 0.9998$ and near-exact parameters (Table 4).

Table 4: Phase-segmented boosting on double-Gompertz synthetic diauxic data

Component	A (true)	A (rec.)	μ (rec.)	λ (rec.)
Phase 1	0.550	0.552	0.350	6.02
Phase 2	0.400	0.398	0.200	26.02

Smooth diauxic: Gompertz-logistic hybrid. For Monod 2-substrate diauxic data where the catabolite repression switch is smooth, the best compact autonomous ODE found is:

$$X'(t) = r\,X\,\ln(K/X)\,(1 + a\,X) \tag{17}$$

The additional term $1 + aX$ represents density-dependent modulation of the Gompertz growth law. For $a > 0$, higher biomass leads to faster effective growth — a possible signature of quorum sensing, cross-feeding, or pH buffering at higher cell densities.

6.3 Model-Free ODE Discovery on Synthetic Diauxic Data

We apply our full pipeline to synthetic Monod 2-substrate diauxic data in a *genuinely model-free* setting: no prior knowledge of the Gompertz, aHPM, or any other form is assumed. Data is generated from:

$$\dot{X} = \left[\mu_1 S_1/(S_1 + K_s) \cdot H(S_1) + \mu_2 S_2/(S_2 + K_s) \cdot (1 - H(S_1))\right] X \tag{18}$$

$$H(S_1) = S_1^n \,/\, (S_1^n + K_d^n) \tag{19}$$

with $\mu_1 = 0.90$, $\mu_2 = 0.25$, $K_d = 0.005$, $n = 10$, observable $X(t)$ only.

Step 1: Derivative fingerprint. Computing $X'(t)$ via Gaussian filter reveals **two distinct peaks** at $t = 4.51$ and $t = 7.64$, with a trough at $t = 5.39$ — the definitive diauxic signature. This detection requires no model assumption.

Step 2: 2nd-order autonomous ODE discovery. Applying STRidge to the library $\Theta = [1, X, X^2, X^3, X', (X')^2, X{\cdot}X', X{\cdot}\ln X, \ldots]$ to discover $X'' = F(X, X')$ produces sparse equations with derivative residuals as low as 0.04, but forward integration fails ($R^2 \ll 0$). This confirms our theoretical result: a compact autonomous ODE cannot capture non-autonomous diauxic dynamics.

Step 3: Direct ODE scan discovers Gompertz structure. Applying the B2 direct scan (no model assumption) — scanning $K \in [X_{\max}, 3X_{\max}]$ and solving $r = \langle X', \Phi\rangle / \|\Phi\|^2$ where $\Phi = X \ln(K/X)$ — the pipeline identifies:

$$\boxed{X' = r\, X\, \ln(K/X), \quad r = 0.395,\ K = 1.540} \tag{20}$$

without assuming Gompertz form a priori. Forward integration gives $R^2 = 0.968$. This is the key demonstration of model-free discovery: the Gompertz ODE emerges as the dominant structure from a generic feature scan.

Why the Gompertz ODE dominates. For smooth Monod kinetics with sharp catabolite repression ($n = 10$, $K_d = 0.005$), the diauxic transition in $X(t)$ is nearly invisible — the parametric Gompertz fits $R^2 = 0.992$. The diauxic structure is visible only in $X'(t)$ (two peaks), not in $X(t)$ itself. This is an important fundamental limitation: model-free discovery from $X(t)$ alone cannot distinguish smooth Monod diauxic from single-phase Gompertz growth.

Table 5: Model-free ODE discovery on synthetic Monod diauxic data

Method	R^2	Notes
Direct Gompertz scan (model-free)	0.968	discovered without prior
Gompertz parametric (baseline)	0.992	known model
2nd-order autonomous ODE	<0	fails: non-autonomous dynamics
Non-autonomous STRidge	<0	overfits derivative space
aHPM (autonomous)	<0	wrong model class

Conclusion. The characterize-first pipeline applied to diauxic data correctly: (1) identifies the two-phase signature in $X'(t)$, (2) rules out autonomous 2nd-order ODEs, (3) discovers the Gompertz ODE as the dominant structure from a feature scan. When the diauxic transition is sharp (double-Gompertz data), phase-segmented boosting recovers both phases exactly (Table 4). When the transition is smooth (Monod data), the derivative fingerprint provides the only reliable diauxic evidence.

7 Grammar Constraint for Symbolic Regression

7.1 From ODE to Grammar Prior

The characterize-first pipeline produces structured information that constrains any downstream symbolic regression search:

1. **Model class**: B3 STRidge identifies the dominant ODE term (e.g., $f \ln f \to$ Gompertz-like). This eliminates oscillatory terms (sin, cos) from the grammar and prioritises exp and ln primitives.
2. **Solution template**: B2 identifies K and r, giving the functional form $f = K \exp(-C \exp(-r(t-t_0)))$ as a starting point. GP then searches for deviations from this template.
3. **Symmetry detection**: parity (even/odd), periodicity, and scaling symmetries further restrict the search. Monotone+saturating data rules out periodic solutions entirely.

7.2 Grammar Constraint in Practice

Once the ODE structure is known, downstream symbolic regression search can be constrained to expression classes consistent with the detected ODE. Concretely:

- Oscillatory terms (sin, cos) are excluded for monotone sigmoidal data.
- The ODE-derived carrying capacity K and rate r provide warm-start initial constants, replacing random initialisation.
- The detected model class (Gompertz, Richards, aHPM) defines the functional skeleton; any remaining free search is over the residual only.

This is a natural direction for future work combining the Python ODE pipeline with a GP back-end.

7.3 Phase-Segmented Symbolic Boosting

For multi-phase growth (diauxic), we propose a phase-segmented boosting approach. Define the diauxic valley t^* as the minimum of X' between the two derivative peaks. Then:

1. Fit G_1 to data on $[0, t^*]$ (phase 1).
2. Compute residual $r(t) = X(t) - G_1(t)$.
3. Fit G_2 to the residual on $[t^*, T]$ (phase 2).

This recovers both Gompertz components with $R^2 = 0.9998$ (Table 4), without any prior assumption that the data contains exactly two phases.

8 Discussion and Future Work

Limitations. The current pipeline handles scalar univariate functions. Extension to multivariate and dynamical systems (PDEs, coupled ODEs) is straightforward in principle but requires adapting the derivative matrix construction. Euler ODEs ($f' = f/x$) with non-constant coefficients are not yet handled.

Weak/integral formulation. For very noisy data (SNR < 5), differentiation fails. Integration against test functions $\langle f', \phi\rangle = -\langle f, \phi'\rangle$ avoids differentiation entirely [10]. This is implemented but not yet benchmarked.

Non-Gompertz data. The B2 scanner assumes Gompertz structure. A general approach would scan over a library of ODE forms (logistic, Richards, power-law) and select the best fit by AIC/BIC. B3 partially addresses this via STRidge model classification.

Full SR pipeline. The complete characterize-first pipeline is:

1. Symmetry detection (parity, periodicity, scaling).
2. Linear ODE inversion → oscillatory/polynomial/exponential class.
3. If residual high: nonlinear ODE detection (B1/B2/B3) → growth model class.
4. Grammar restriction based on detected class.
5. Constrained GP or exhaustive search on restricted grammar.
6. Symbolic boosting on residuals.

9 Conclusion

We have demonstrated that ODE inversion is a practical and effective pre-processing step for symbolic regression. On bacterial growth data, the direct Gompertz ODE scanning (B2) recovers exact model parameters on synthetic data and achieves $R^2 \geq 0.992$ on real noisy measurements, matching the best parametric fits. The method is noise-robust (working to $\sim$20% additive noise with Gaussian derivative filters and GCV bandwidth selection), computationally cheap (seconds on 200-point time series), and produces interpretable output: the governing ODE and its analytical solution form.

The characterize-first paradigm inverts the usual SR workflow: instead of searching blindly and hoping to find the right expression, we *characterize what class the expression must belong to* before searching. This transforms an intractable combinatorial search into a well-posed algebraic problem.

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